The Kingdom's Climate Pulse: Mining, Clustering, and Modeling Hourly Weather Data

A PROJECT REPORT

Submitted by

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MATH 506

Fundamentals of Data Science

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Introduction

Weather Datasets are always huge in size as they are collected in a very frequent manner. Intelligently working on them and understanding them might give us fruitful knowledge about complex weather conditions of our landscape.

In this project, we have taken a weather dataset from a very legitimate data source called the National Oceanic and Atmospheric Administration. This dataset gives us meteorological information about air stations that are in the Kingdom of Saudi Arabia. We will carefully prepare the data and then derive useful insights from it that is beneficial for the public welfare.

We were able to perform good clustering and classification models to the dataset after careful cleaning methods. Frequent pattern mining were also performed in order to understand the most recurring patterns in the dataset.

Data Description

The dataset has more than 9,00,000 entries and 35 columns of both numerical and categorical features. Obviously, there are also missing values. Dataset is collected from https://www.noaa.gov/.

Data Cleaning and Preprocessing

To ensure model effective, the raw sensor data from the dataset underwent a detailed cleaning pipeline. First, preprocessing involved converting timestamps and removing administrative artifacts. Then the application of physical constraints to filter impossible outliers. An important step was addressing the high missingness in variables like CLOUD_CEILING. Therefore, station selection method was implemented to retain only the **25 high-quality stations** that have <40% missing data in key fields. For these valid stations, like Jeddah, missing continuous values were imputed using station-specific means to maintain local weather values, while binary values were changed to distinguish between "missing data" and "clean" nulls.

Feature Engineering

Some prominent feature engineering involve:

```python
import numpy as np

# 1) Temperature-dewpoint spread (°C)
df_high_quality['TEMP_DEW_SPREAD'] = (
    df_high_quality['AIR_TEMPERATURE'] - df_high_quality['AIR_TEMPERATURE_DEW_POINT']
)

# 2) Approximate Relative Humidity (%)
# Using Magnus formula; safe for typical surface temps.
T = df_high_quality['AIR_TEMPERATURE']
Td = df_high_quality['AIR_TEMPERATURE_DEW_POINT']

# Avoid weird NaNs or overflows by clipping temps to a reasonable range
T_clip = T.clip(-80, 60)
Td_clip = Td.clip(-80, 60)

es = 6.112 * np.exp((17.67 * T_clip) / (T_clip + 243.5))      # saturation vapor pressure
e  = 6.112 * np.exp((17.67 * Td_clip) / (Td_clip + 243.5))    # actual vapor pressure

RH = 100 * (e / es)
df_high_quality['RELATIVE_HUMIDITY'] = RH.clip(0, 100)
```

Pattern Mining & Association Analysis

We set the minimum support to be 50% percent and the minimum confidence to be 50% too. We ended up creating few strong associations rules, with a ver high confidence. We were able to derive key patterns from doing this. As discussed in our MATH 506 lectures, we wer fully able to understand how Apriori Algorithm worked and how useful it is when applied to such complex pattern dense datasets.

Modelling Approach

The methodology taken for the study was hybrid unsupervised and supervised to identify and validate meteorological regimes.

6.1 Unsupervised Regime Discovery

Used K-means Clustering algorithm to find the prominent cluster groups that were hiding in the dataset. We randomly chose the k to be 4, but it turned out that it was a good choice, because the dataset actually contained 4 prominent weather patterns.

6.2 Supervised Pattern Validation

We then employed RandomForest Classifier in order to check if the model was able to learn the meteorological patterns underlying the data. We were stunned to see the model performance.

Model Evaluation

The 4 clusters that came out from K-Means Clustering Algorithm were very realisitc. When trained the RandomForest Algorithm performed well on understanding the patterns and were able to predict the weather labels correctly.

```
Classification report:
              precision    recall  f1-score   support

           0       0.99      1.00      0.99     14913
           1       1.00      0.99      1.00      4276
           2       0.99      0.99      0.99     11911
           3       1.00      1.00      1.00      2016

    accuracy                           0.99     33116
   macro avg       1.00      1.00      1.00     33116
weighted avg       0.99      0.99      0.99     33116


Confusion matrix:
[[14844     8    61     0]
 [   10  4253    13     0]
 [   76    11 11824     0]
 [    0     0     0  2016]]
```

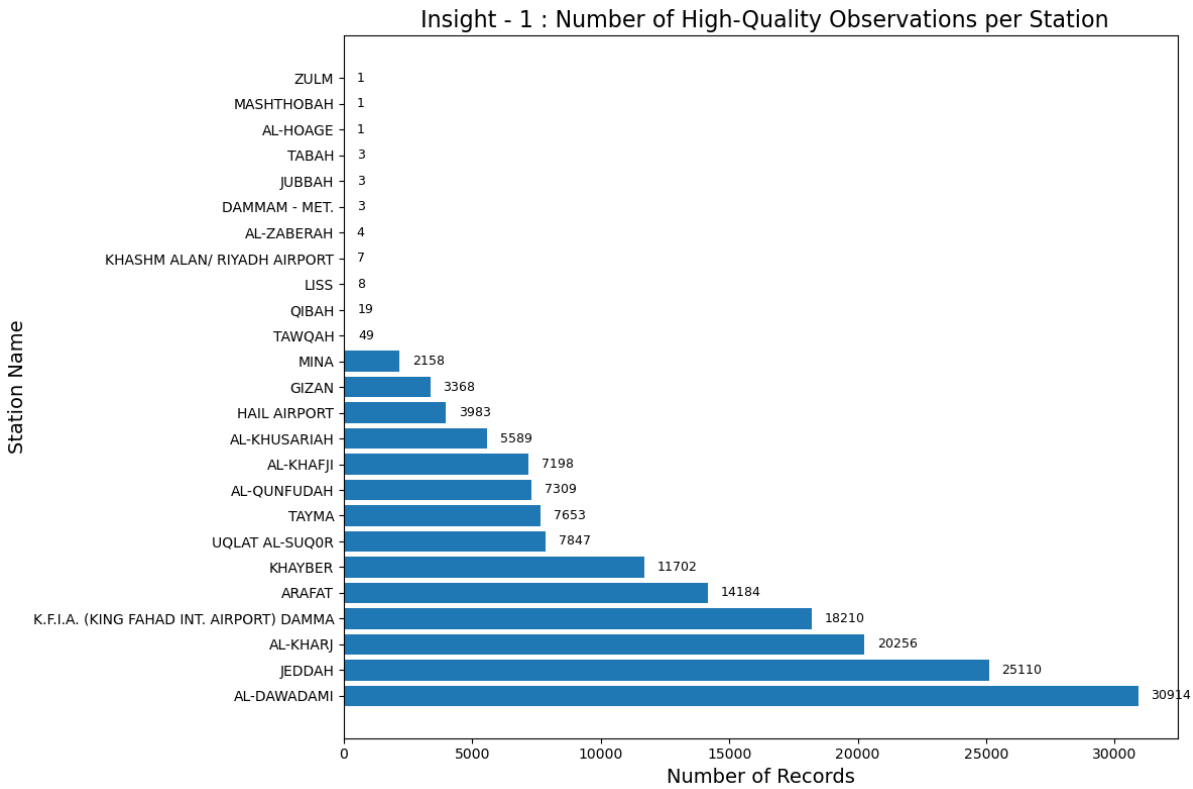
Key Insights

We were able to derive these very useful and important insights from the dataset after al the efforts mentioned above.

8.1 Insight 1: Important Airports Have Rich Data

Out of all 88 Air Stations we saw at the raw dataset, 2 stations were very data rich. The below plot supports this fact.

- **Interpretation:** Al Dawadami and Jeddah airports being densely populated airports in the whole kingdom, prove to have rich data collection sensors, very few missing values compared to other air stations.

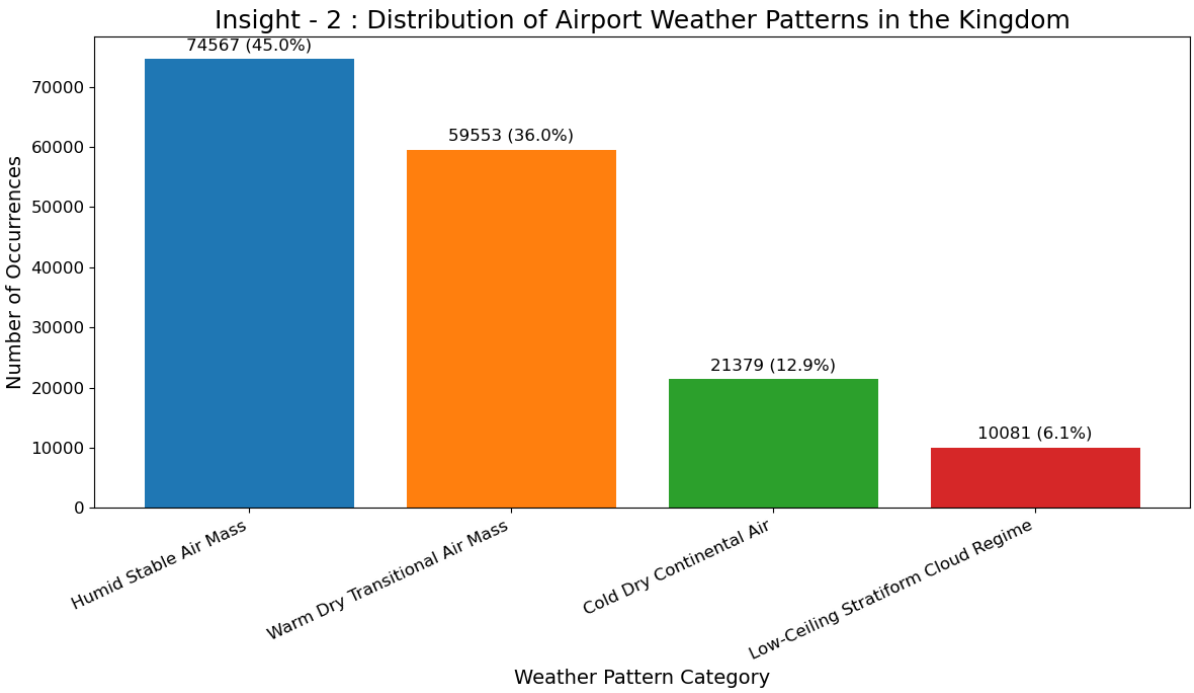


8.2 Insight 2: 4 Prominent Weather Patterns

Out of the 4 prominent weather patterns that are often recorded in the Kingdom's Air Stations, Humid Stable Air is the most occurring weather. It accounts to 45% of the whole cleaned dataset.

- **Interpretation:** This does not support the stereotype of desert climate. It indicates that a plurality of the Kingdom's major airports is situated in zones heavily influenced by

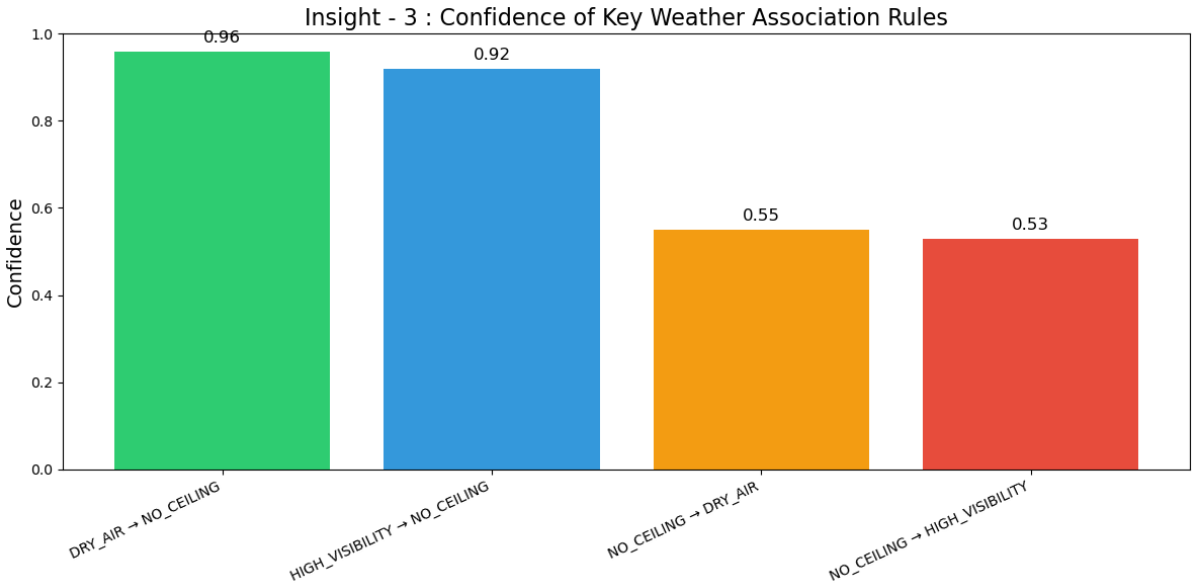
maritime moisture advection (Red Sea/Arabian Gulf), which moderates extreme dryness more often than purely continental regimes (which appeared in only 21,379 instances).



8.3 Insight 3: Deterministic Physical Associations

Association Rule Mining (Apriori) quantified the probabilistic relationships between meteorological variables. The strongest discovered rule was $\{DRY_AIR\} \rightarrow \{NO_CEILING\}$with a confidence of **0.96**.

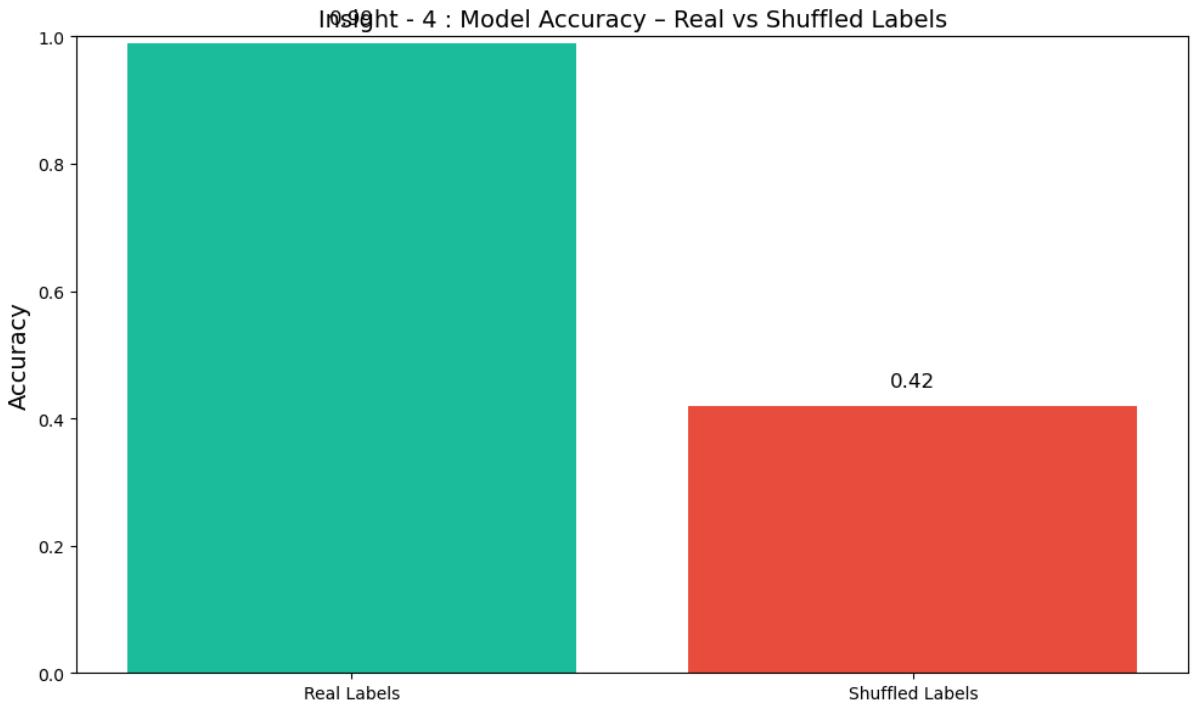
- **Interpretation:** This near-perfect correlation establishes a deterministic link: when relative humidity drops below 40%, the probability of clear skies is 96%. Conversely, the inverse rule $\{NO_CEILING\} \rightarrow \{DRY_AIR\} \rightarrow$ had a confidence of only 55%, revealing that clear skies can frequently occur during humid conditions, but dry air almost never sustains low cloud cover.



8.4 Insight 4: Validity of Learned Regimes (Signal vs. Noise)

We compared two RandomForest Classifiers, one with real labels while the other one with shuffled classes (same rows but classes shuffled). This is done in order to check if the model trained on real classes were legit and not just memorising.

- **Real Labels:** 99% Accuracy.
- **Shuffled Labels:** 42% Accuracy.
- **Interpretation:** Therefore, it is very clear that the model was very genuinely learning the patterns, if it was just memorising, then the shuffled label accuracy would also be too good.

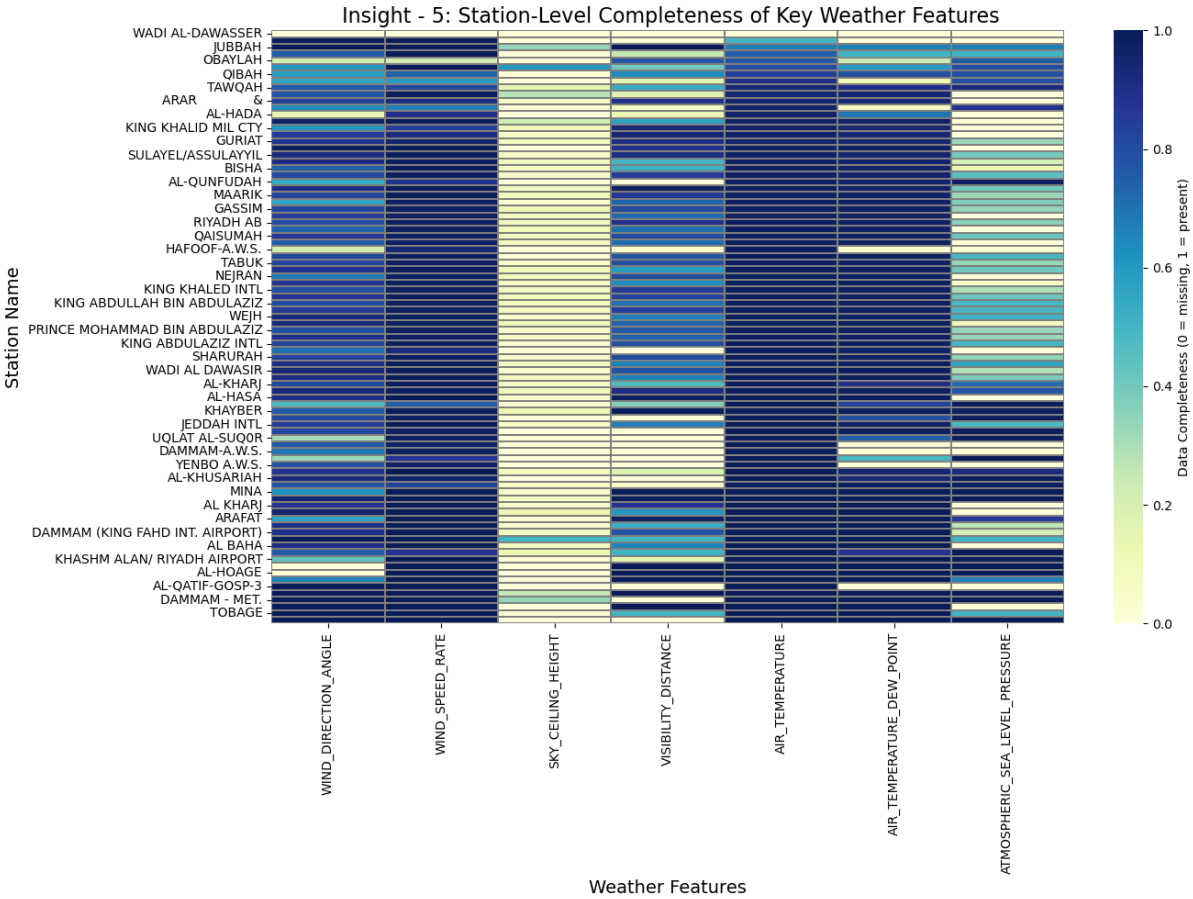


8.5 Insight 5: Systemic Missingness in Cloud Data

Sky Ceiling sensors are very important in aviation because almost tells us if it is safe to take of or not.

The below heatmap shows us that more than 90% of the air stations in Saudi Arabia does not have a very strong SKY_CEILING sensor.

- **Interpretation:** In KSA sky ceiling sensors are not widely installed. Sky Ceiling sensors are very important in aviation because almost tells us if it is safe to take of or not. The below heatmap shows us that more than 90% of the air stations in Saudi Arabia does not have a very strong SKY_CEILING sensor.



Conclusion

This project successfully demonstrates how a real-world data science project would work. We also understood that model training is not the hardest part in data science but rather the parts like data collection and cleaning takes a lot of effort and time. This

project built a very strong foundation for us to work with large real-world datasets in the future.